Finite Potential Well

$$V(x) = \begin{cases} V_0, & x < -\frac{L}{2} \quad \text{(Region III)} \\ 0, & -\frac{L}{2} \le x \le \frac{L}{2} \quad \text{(Region I)} \\ V_0, & x > \frac{L}{2} \quad \text{(Region II)} \end{cases}$$

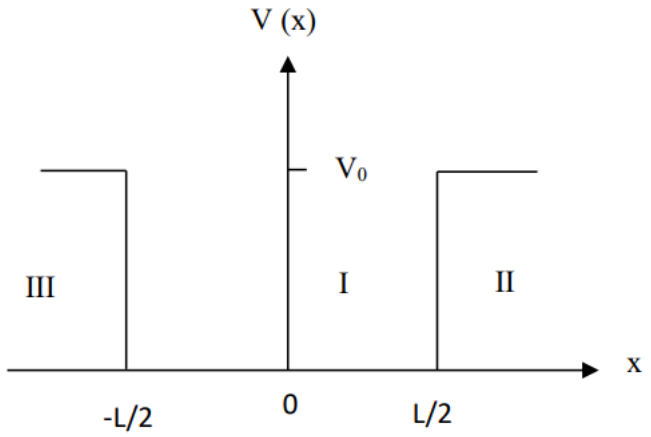
$$0 < E < V_0$$

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + V_0\psi(x) = E\psi(x)$$

or

$$\psi''(x) = \alpha^2\psi(x)$$

$$\psi_{III}(x) = Fe^{-\alpha x} + Ge^{\alpha x}$$



$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) = E\psi(x)$$

$$\psi''(x) = -k^2\psi(x)$$

$$\psi_I(x) = A\sin kx + B\cos kx,$$

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + V_0\psi(x) = E\psi(x)$$

or

$$\psi''(x) = \alpha^2\psi(x)$$

where

$$\alpha^2 = \frac{2m(V_0 - E)}{\hbar^2} > 0$$

$$\psi_{II}(x) = Ce^{-\alpha x} + De^{\alpha x}$$

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